FLUTICASONE

Ritonavir — Cushing's

Rinse mouth

CHEMICAL NAME S-(fluoromethyl) 6α,9-difluoro-11β,17-dihydroxy-16α-methyl-3-oxoandrosta-1,4-diene-17β-carbothioate (propionate or furoate ester)

CLASS Potent inhaled/intranasal corticosteroid

TRADE NAMES Flixotide · Flovent · Flixonase / Avamys (nasal) · Seretide (with salmeterol)

THERAPEUTIC USES / INDICATIONS

- Asthma maintenance (preventer).
- COPD (with LABA ± LAMA).
- Allergic rhinitis, nasal polyps (nasal spray).
- Eczema (topical cream).

PHARMACODYNAMICS

- High-affinity **glucocorticoid receptor** agonist → potent local anti-inflammatory action.

ADMINISTRATION

- **Inhaler** twice daily (propionate) or once daily (furoate in combinations); spacer with pMDI.
- Rinse mouth and spit after each use.
- Nasal: aim away from septum.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Oral thrush, hoarseness.
- Headache, cough.
- Nasal: epistaxis, dryness.

ADVERSE EFFECTS (SERIOUS)

- Adrenal suppression (high doses, esp. children).
- Growth suppression; ↓ bone density.
- Pneumonia in COPD.
- Cataract, glaucoma.

DOSAGES

- **Adult asthma:** 100–500 mcg twice daily (propionate).
- **Children 4–16 yr:** 50–100 mcg twice daily (max 200 mcg twice daily).
- **Nasal (propionate):** 2 sprays each nostril once daily; furoate 2 sprays daily.
- Fluticasone is ~2× potency of beclometasone (Clenil).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** TB, untreated infections.

KEY INTERACTIONS

- Ritonavir and cobicistat: markedly ↑ fluticasone → Cushing's syndrome and adrenal suppression — avoid.
- **Ketoconazole, itraconazole:** ↑ levels.

PHARMACOKINETICS

- **Onset:** days; full effect 1–2 weeks.
- Negligible oral bioavailability (<1%); metabolised by CYP3A4.
- Excreted in faeces. **t½:** ~8–14 h (inhaled).

NURSING MANAGEMENT

- Check technique, adherence.
- Monitor children's growth.
- Check for HIV protease inhibitors.
- Inspect mouth.

PATIENT EDUCATION

- Rinse mouth after each use.
- Use daily, even when well.
- Tell doctors if you start HIV medicines.
- Nasal spray takes several days to work.

KEY POINTS

- Potent ICS.
- Rinse mouth.
- Ritonavir interaction → Cushing's.
- Monitor children's growth on high doses.